

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Zoology**Paper: ZOB-505 — Environmental Biology and Systematics**

Time: Three hours

Full Marks: 70

Note: Answer four questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A**(Environmental Biology) (Marks: 45)****Question 1.** Answer the following parts:

- (a) Write whether the following statements are True or False. Justify your answer in 2-3 sentences each: [2 × 5 = 10]

- (i) The oceans are biological deserts.
- (ii) The transport of oxygen from lungs to other parts of the body is facilitated by carboxyhaemoglobin.
- (iii) Stratospheric ozone forms naturally and helps shield us from ultraviolet radiation of sunlight.
- (iv) Hexavalent chromium is not a pollutant and is beneficial to humans.
- (v) Invasive exotic species are the main threat to biodiversity.

- (b) Define the following terms: [1 × 5 = 5]

- (i) Keystone species
- (ii) Gross primary productivity
- (iii) Ecesis
- (iv) Fecundity rate
- (v) Pyramid of energy

Question 2. Write an essay on the aquatic ecosystems. [10]

Question 3. What is solid waste pollution? Describe various methods of solid waste management along with their advantages. [10]

Question 4. Write an essay on the nature, structure, and attributes of a biological community. [10]

Question 5. Write short notes on the following: [2.5 × 4 = 10]

- (a) Food web
- (b) Energy flow in ecosystem
- (c) Positive interactions
- (d) Biodiversity conservation

SECTION – B
(Systematics) (Marks: 25)

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False. Justify your answer in 2-3 sentences each: [2 × 2 = 4]

- (i) Tribe is a lower taxonomic category that comes between genus and species.
- (ii) DNA-DNA hybridization is a more reliable method to establish genetic similarity between two species than comparing their nucleotide sequences of certain common genes.

(b) Define the following terms: [1 × 3 = 3]

- (i) Taxon
- (ii) Tautonym
- (iii) Binomial nomenclature

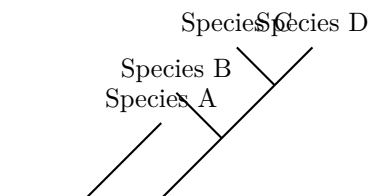
Question 2. Discuss the following: [4.5 × 2 = 9]

- (a) Molecular taxonomy
- (b) Taxonomic characters

Question 3. What is a species? Discuss different types of species concepts and specify which one of the concepts finds maximum acceptance from biologists, giving the reasons for its acceptance. [9]

Question 4. Write notes on the following: [1.5 × 6 = 9]

- (a) DNA barcoding
- (b) Polytypic species
- (c) Phylogenetic tree



- (d) Alpha taxonomy
- (e) Genus
- (f) Category

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